기출 최적제어 2014년 중간고사 1번 문제 4.12

교재181p 노트25p 4.2.X 단원 케이스1-2

$J(x)=\int_0^1 (x\cdot\dot x+\ddot x^2)\,dt$ $x(t_0)=x(0)=0$ $x(t_f)=x(1)=2$ $t_0=0$

$\dot x(t_0)=\dot x(0)=1$ $\dot x(t_f)=\dot x(1)=4$ $t_f=1$

(i) EL식) $\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot x}\right)-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot x}\right)+\frac{\partial g}{\partial x}=0$.

$\frac{\partial g}{\partial x}=\dot x$, $\frac{\partial g}{\partial \dot x}=x \longrightarrow \frac{d}{dt}x=\dot x$, $\frac{\partial g}{\partial \ddot x}=2\ddot x \longrightarrow \frac{d^2}{dt^2}2\ddot x=2\ddddot x$.

EL식에 위 조건을 대입하면) $2\ddddot x-\dot x+\dot x=0$, $\ddddot x=0$, $\dddot x=$상수, $\ddot x=$일차식, $\dot x=$이차식, $x=$삼차식.

$x(t)=at^3+bt^2+ct+d$ $\longrightarrow$ BC조건 $x(0)=0$, $\dot x(0)=1$, $x(1)=2$, $\dot x(1)=4$ $\longrightarrow$ $a=1$, $b=0$, $c=1$, $d=0$ $\longrightarrow$ ∴

$\dot x(t)=3at^2+2bt+c$

$x=t^3+t$	$0\sim2$
$\dot x=3t^2+1$	$1\sim4$
$\ddot x=6t$	$0\sim6$
$\dddot x=6$	6
$\ddddot x=0$	0

$t_0=0$, $t_f=1$

(ii) 횡단식 ⇒ 없음.

기출 최적제어 2013년 중간고사 3번 문제 4.14

교재 181p 노트 32p 4.2.X 단원 케이스4

$J(x)=\int_0^{t_f}\frac{\sqrt{1+\dot x^2}}{x}\,dt$ $t_0=0$, $x(t_0)=0$

$t_f=$자유, $x(t_f)=\theta(t_f)=t_f-5$. (단, 관계성문제).

$x(t_f)=$자유.

(i) EL식) $\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot x}\right)=0$, $\boxed{t=t_0\sim t_f}$, $\frac{\partial g}{\partial x}=-\frac{\sqrt{1+\dot x^2}}{x^2}$, $\frac{\partial g}{\partial \dot x}=\frac{1}{x}\cdot\frac{2\dot x}{2\sqrt{1+\dot x^2}}$, $\frac{d}{dt}\left(\frac{\dot x}{x\sqrt{1+\dot x^2}}\right)=?$

⇒ skip! ⇒ $\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot x}\right)=0 \longrightarrow x\cdot\ddot x+\dot x^2=-1=\frac{d}{dt}(x\cdot\dot x)$ ⇒ 적분. $\boxed{x\dot x=-t+C_1}$

⇓ 적분.

$\frac{x^2}{2}=-\frac{t^2}{2}+C_1t+\cancel{C_2}$

(ii) 횡단식) $\dot x(t_f)=-1$, $t_f^2-(5+C_1)t_f+\frac{25}{2}=0$, $x^2=-t^2+2C_1t$, $x(t_f)=t_f-5$, $\dot x(t_f)=-1$ $\longrightarrow$ $\boxed{4C_1t_f=2t_f^2+C_1^2}$.

BC조건으로 $C_2=0$

∴

$C=5$, $t_f=5\pm\frac{5}{\sqrt2}$ $\longrightarrow$ 선택.

$C=5(\sqrt2+1)$, $t_f=\frac{5}{\sqrt2}$

$C=-5(\sqrt2-1)$, $t_f=-\frac{5}{\sqrt2}$

∴ $\sqrt{x^2}=\sqrt{-t^2+10t}=x$ $t_0=0$, $t_f=5\pm\frac{5}{\sqrt2}$

$x(t_0)=0$, $x(t_f)=\pm\frac{5}{\sqrt2}$

//

중간고사 최적궤도제어

2014 년 가을학기

O-L $\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot x}\right)+\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot x}\right)=0$

⇓

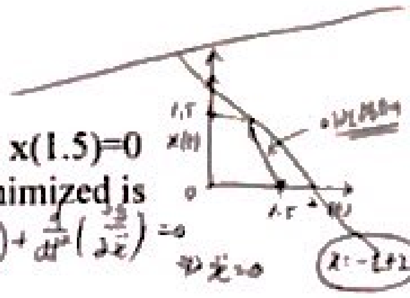
$\ddddot{x}=0$

$x(t)=c_1t^3+c_2t^2+c_3t+c_4$

1. Find the equation of the curve that is an extremal for the functional

$$J(x)=\int_0^1\left[x(t)\dot x(t)+\ddot x^2(t)\right]dt$$

The boundary conditions are $x(0)=0,\ x(1)=2,\ \dot x(0)=1,\ \dot x(1)=4$.



2. Find the shortest piecewise-smooth curve joining the points x(0)=1.5 and x(1.5)=0 which intersects the line x(t) = -t+2 at one point. The functional to be minimized is

$$J(x)=\int_0^{1.5}\left[1+\dot x^2(t)\right]^{1/2}dt$$

O-L eqn' $\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot x}\right)+\frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \ddot x}\right)=0$ ⇒ $\ddot x=0$

Corner condition

$\left[g+\frac{\partial g}{\partial \dot x}(\dot\theta-\dot x)\right]_{t=t_1^-}=\left[g+\frac{\partial g}{\partial \dot x}(\dot\theta-\dot x)\right]_{t=t_1^+}$

⇓

$(1+\dot x^2)^{-\frac12}(1-\dot x)\big|_{t_1^-}$

$=(1+\dot x^2)^{\frac12}(1-\dot x)\big|_{t_1^+}$

3. The system

$$\dot x_1(t)=x_2(t)$$
$$\dot x_2(t)=-x_2(t)+u(t)$$

The system is to be transferred from to $\mathbf{x}(0)$ to the line $x_1(t)+5x_2(t)=15$.

$\vec x(0)=0$

The following performance index is to be minimized

$$J(u)=\int_{t_0}^{t_f}\frac{1}{2}u^2(t)dt$$

The admissible states and control are not bounded. Find the necessary conditions that must be satisfied for optimal control. Solve the problem to get an optimal trajectory.

4. The system

LQR tracking

$$\dot x_1(t)=x_2(t)$$
$$\dot x_2(t)=2x_1(t)-x_2(t)+u(t)$$

is to be controlled to minimize the performance index

$$J(u)=\left[x_1(t_f)-1\right]^2+\int_0^{t_f}\left\{\left[x_1(t)-1\right]^2+0.0025u^2(t)\right\}dt$$

The final time t_f is specified, $\mathbf{x}(t_f)$ are free, and the admissible states and controls are not bounded. Find the optimal control law by using the LQR.

$H=g+\lambda f$

최적궤도제어 (중간고사)

2013 년 가을학기

1. Derive necessary conditions to make the performance measure extremum.

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), \ddot{x}(t), t)\,dt$$ $\delta J = 0$

t_0 and t_f are fixed and $x(t_0), x(t_f), \dot{x}(t_0), \dot{x}(t_f)$ are given

2. Find the equation of the curve that is an extremal for the functional

$$J(x) = \int_0^{t_f} \left[t\,\dot{x}(t) + \dot{x}^2(t) \right] dt$$

$t_f = 1$ and $x(0) = 1$, $x(1) = 2.75$.

3. Find the extremal and final time of

$$J(x) = \int_0^{t_f} \frac{\sqrt{1+\dot{x}^2}}{x} dt$$

Subject to $x(0) = 0$ and $x(t_f)$ must lie on the line $\theta(t) = t - 5$. The final time is free.

4. $\dot{x}(t) = -ax(t) + bu(t)$, $a>0$, $b>0$, a and b are constant.

$x(0) = 0$, $x(t_f) = 15$, t_f is specified.

Determine control $u(t)$ to minimize

$$J(x) = \frac{1}{2}\int_0^{t_f} u^2(t)\ dt$$

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시 험 답 안 지

과목명	최적제어제어 Mid-term

~~공~~ 대학 학과(계열) 학년

학번 2015311294 성명 최다솜

(20 년 월 일 실시)

성 적	75

1. $J(x) = \int_0^{\frac{\pi}{2}} [\dot{x}_1^2(t) + \dot{x}_2^2(t) + 2x_1(t)x_2(t)]\,dt$

$x_1(0)=0,\ x_1(\frac{\pi}{2})=1 \qquad x_2(0)=0,\ x_2(\frac{\pi}{2})=1$

15/25

Euler-Lagrange equation

$\frac{\partial g}{\partial x} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) = 0$

$\frac{\partial g}{\partial x_1} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}_1}\right) = 0 \qquad \frac{\partial g}{\partial x_2} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}_2}\right) = 0$

$\Rightarrow 2x_2 - \frac{d}{dt}(2\dot{x}_1) = 0 \qquad \Rightarrow 2x_1 - \frac{d}{dt}(2\dot{x}_2) = 0$

$x_2 = \ddot{x}_1 \qquad x_1 = \ddot{x}_2$

$x_1 = \ddddot{x}_1 \qquad x_1 = \sin t$

$\lambda^4 - 1 = 0 \qquad \ddot{x}_2 = \sin t$

$\lambda^4 = 1 \qquad \dot{x}_2 = -\cos t + c_1$

$\lambda^2 = 1 \text{ or } -1 \qquad x_2 = -\sin t + c_1 t + c_2$

$\lambda = \pm 1 \text{ or } \pm j$

$x_1(t) = c_1\cos t + c_2 \sin t$

$x_1(0) = 0 = c_1 \qquad x_2(0) = c_2 = 0$

$x_1(\frac{\pi}{2}) = 1 = c_2 \qquad x_2(\frac{\pi}{2}) = 1 = -1 + \frac{\pi}{2}c_1$

$\frac{\pi}{2}c_1 = 2 \qquad c_1 = \frac{4}{\pi}$

$\therefore x_1^*(t) = \sin t$

$x_2^*(t) = -\sin t + \frac{4}{\pi}t$

$\rightarrow x_1(t) = c_1e^{-t} + c_2e^{t} + c_3\cos t + c_4 \sin t$

$x_2(t) = \quad '' \quad + (-c_3\cos t - c_4 \sin t)$

$c_3 = c_4 = 0 \qquad \Rightarrow \therefore x_1(t) = x_2(t) = \frac{\sinh(t)}{\sinh(\frac{\pi}{2})}$

55
10
25
25
75

연세대학교

앞면에서 계속

2. $J = \int_0^1 [\dot{x}^2 + t^2] dt$

$x(0)=0, \quad x(1)=0$

$z = \int_0^1 x^2 dt = 2$ ~~and constraint~~

$z(t_0)=0, \quad z(t_f)=2$

$\dot{z} = x^2 = F$

let augmented

$g_a = g + \lambda(x^2 - \dot{z})$

Euler-Lagrange equation

$\frac{\partial g}{\partial x} - \frac{d}{dt}\left[\frac{\partial g}{\partial \dot{x}} + \lambda \frac{\partial f}{\partial \dot{x}}\right] + \lambda\left(\frac{\partial f}{\partial x}\right) = 0$

$\Rightarrow \quad 0 - \frac{d}{dt}[2\dot{x} + 0] + \lambda(2x) = 0$

$\frac{d}{dt}(\lambda \frac{\partial f}{\partial \dot{z}}) = 0$

$\frac{\partial g}{\partial z} - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{z}}\right) = 0 \Rightarrow \lambda = const$

$g_a = \dot{x}^2 + 2t + \lambda(x^2 - \dot{z})$

i) $\lambda = c \geq 0$

$\cancel{2}\ddot{x} = \cancel{2}\lambda x$

$\ddot{x} = \lambda x$

let $x = e^{at} \quad \dot{x} = ae^{at} \quad \ddot{x} = a^2 e^{at}$

$a^2 = \lambda \quad a = \pm\sqrt{\lambda}$

$x = c_1 \cos\sqrt{\lambda}t + c_2 \sin\sqrt{\lambda}t.$

$\longrightarrow x = a_1 e^{\sqrt{c}t} + a_2 e^{\sqrt{c}t}$

$\uparrow$ B.C ~~$x_1, x_2 \Rightarrow$~~ $a_1 = a_2 = 0$ (trivial)

ii) $\lambda = c < 0$

$\ddot{x} + |c|x = 0$

$x = b_1 \cos(\sqrt{|c|}t) + b_2 \sin(\sqrt{|c|}t)$

$= \pm 2 \sin(n\pi t), \quad n = 1, 2, 3 \cdots$

시 험 답 안 지

과목명	최적궤도제어

대학 학과(계열) 학년

학번 2015311294 성명 최다솜

(~~20~~ 2015 년 10 월 21 일 실시)

성 적	

3. general optimal problem $H = g + \vec{\lambda}^T \vec{x}$, $H = H(\vec{x}, \vec{u}, \vec{\lambda}, t)$

augmented functional을 아래와 같이 정해준다.

$$\cancel{J_a(x) = h(\vec{x}(t_f), t_f) + \vec{\nu}^T \vec{m} + \int_{t_0}^{t_f+\delta t_f} (H - \vec{\lambda}^T \dot{\vec{x}})\, dt}$$

$$\delta J_a(x) = \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \frac{\partial h}{\partial t_f} \delta t_f + \delta \vec{\nu}^T \vec{m}^{\,0} + \vec{\nu}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \vec{\nu}^T \frac{\partial \vec{m}}{\partial t_f} \delta t_f$$

$$+ \int_{t_0}^{t_f} \left(\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\lambda}} \delta \vec{\lambda} - \delta \vec{\lambda}^T \dot{\vec{x}} - \vec{\lambda}^T \delta \dot{\vec{x}} \right) dt$$

$$+ \int_{t_f}^{t_f+\delta t_f} (H - \vec{\lambda}^T \dot{\vec{x}})\, dt$$

① (* 부분적분 이용 $\int -\vec{\lambda}^T \delta \dot{\vec{x}}\, dt = -\vec{\lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f} + \int \dot{\vec{\lambda}}^T \delta \vec{x}\, dt$)

② ($\int_{t_f}^{t_f+\delta t_f} (H - \vec{\lambda}^T \dot{\vec{x}})\, dt \simeq (H - \vec{\lambda}^T \dot{\vec{x}}) \delta t_f$ 넓이로 생각할수있음.

③ $\delta \vec{x}(t_f) = \delta \vec{x}_f - \dot{\vec{x}}(t_f) \delta t_f = \delta \vec{x}_f - \dot{\vec{x}}^*(t_f) \delta t_f$ 이용 ①에넣으면,

$$-\vec{\lambda}^T \delta \vec{x}(t_f) = -\vec{\lambda}^T (\delta \vec{x}_f - \dot{\vec{x}}^*(t_f) \delta t_f)$$

①,②,③ 이용 $\delta J_a(x)$를 계속 정리하면

$$\delta J_a(x) = \underbrace{\left(\frac{\partial h}{\partial \vec{x}(t_f)} + \vec{\nu}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} - \vec{\lambda}^T \right)}_{(\mathrm{IV}-1)} \delta \vec{x}_f + \underbrace{\left(\frac{\partial h}{\partial t_f} + \vec{\nu}^T \frac{\partial \vec{m}}{\partial t_f} + \cancel{\vec{\lambda}^T \dot{\vec{x}}} + H - \cancel{\vec{\lambda}^T \dot{\vec{x}}} \right)}_{(\mathrm{IV}-2)} \delta t_f$$

$$+ \int_{t_0}^{t_f} \left[\underbrace{\left(\frac{\partial H}{\partial \vec{u}} \right)}_{(\mathrm{I})} \delta \vec{u} + \underbrace{\left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right)}_{(\mathrm{II})} \delta \vec{x} + \delta \vec{\lambda}^T \left(\left(\frac{\partial H}{\partial \vec{\lambda}} \right)^T - \dot{\vec{x}} \right) \right] dt$$

$\delta J_a = 0$ by fundamental theorem

∴ Necessary conditions

(I) optimality $\frac{\partial H}{\partial \vec{u}} = 0$

(II) costate eqn $\dot{\vec{\lambda}} = -\left(\frac{\partial H}{\partial \vec{x}} \right)^T$

(III) state eqn $\dot{\vec{x}} = \left(\frac{\partial H}{\partial \vec{\lambda}} \right)^T$

(IV) Boundary condition at $t = t_f$

$\vec{\lambda} = \left(\frac{\partial h}{\partial \vec{x}(t_f)} + \vec{\nu}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} \right)^T$

$H = -\left(\frac{\partial h}{\partial t_f} + \vec{\nu}^T \frac{\partial \vec{m}}{\partial t_f} \right)$

(V) $\vec{x}(t_0) \Rightarrow$ given

시험답안지

과목명	최적궤도제어

대학 학과(계열) 학년

학번 2015제1294 성명 최다솜

(2015 년 10 월 21 일 실시)

성적

LQR method

$\dot{\vec{x}} = A\vec{x} + B\vec{u}$

$J = \frac{1}{2}\int (\vec{x}^T Q \vec{x} + \vec{u}^T R \vec{u})\, dt$

Let Hamiltonian, $H = g + \vec{\lambda}^T \dot{\vec{x}} = \frac{1}{2}(\vec{x}^T Q \vec{x} + \vec{u}^T R \vec{u}) + \vec{\lambda}^T \dot{\vec{x}}$

$+ \vec{\lambda}^T (A\vec{x} + B\vec{u})$

$\frac{\partial H}{\partial \vec{u}} = R\vec{u} + B^T \vec{\lambda} \Rightarrow \vec{u} = -R^{-1} B^T \vec{\lambda}$

$\frac{\partial H}{\partial \vec{x}} = \cancel{Q\vec{x} + A^T \vec{\lambda}} \Rightarrow \dot{\vec{\lambda}} = Q\vec{x} + A^T \vec{\lambda}$

Let $\vec{\lambda} = K\vec{x}$

$\vec{u} = -R^{-1} B^T K \vec{x}$

$\dot{\vec{\lambda}} = \dot{K}\vec{x} + K\dot{\vec{x}}$

$Q\vec{x} + A^T \vec{\lambda} = \dot{K}\vec{x} + K(A\vec{x} - BR^{-1}B^T K \vec{x})$

$Q\vec{x} + A^T K \vec{x} = \dot{K}\vec{x} + KA\vec{x} - KBR^{-1}B^T K \vec{x}$

$\Rightarrow \dot{K} = KA - KBR^{-1}B^T K + A^T K + Q$ // ~~Riccati equation~~

A, B, Q, R이 다 time-invariant면

$\boxed{Q + 2KA - KBR^{-1}B^T K = 0}$ // ~~Algebraic Riccati equation~~

앞면에서 계속

Example 5-2-2

4. $\dot{x}_1 = x_2$

$\dot{x}_2 = 2x_1 - x_2 + u$

$$J(u) = \int_0^T \{ \underline{x_1^2(t) + \frac{1}{2} x_2^2(t) + \frac{1}{4} u^2(t)} \} dt$$

LQR method

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u$$

$$[x_1 \ x_2] \begin{bmatrix} Q \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + \frac{1}{2} x_2^2 \Rightarrow Q = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

같은방법으로 $R = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

A, B, Q, R을 아므로 다음쪽의 Algebraic Riccati equation 에서

K를 구할수 있음.

$\vec{\lambda} = K\vec{x}$ 이므로 $\vec{\lambda}$를 구할수 있고, 이를 통해

$\vec{u} = -R^{-1}B^T\vec{\lambda} = -R^{-1}B^TK\vec{x}$ 이므로

optimal control $u(t)$를 구할수 있다.

2019-2 최적제어 중간고사

1. ex 4.3-3 (P. 149)

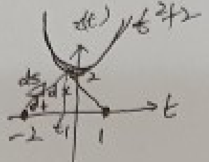
2. piecewise-Smooth extremals

3. Optimal control

Minimize control

4. (LQR) ex 5.2-3 (p. 222)

2019 중간고사.



$d s^2 \approx dt^2 + dx^2$ $\quad$ $\min \int ds$ $\quad$ $\Rightarrow J = \int_{-2}^{1} \sqrt{1+\dot{x}^2}\, dt = \int_{-2}^{t_1^-} \sqrt{1+\dot{x}^2}\, dt + \int_{t_1^+}^{1} \sqrt{1+\dot{x}^2}\, dt$

$ds = \sqrt{1+\left(\frac{dx}{dt}\right)^2}\, dt$

2. piecewise

E-L $\Rightarrow$ $\ddot{x} = 0$, $\dot{x} = c_1$, $x(t) = c_1 t + c_2$ $\quad (-2 \le t \le t_1)$

$x(t) = c_3 t + c_4$ $\quad (t_1 \le t \le 1)$

$$\left.\begin{aligned} x(-2) &= 0 = -2c_1 + c_2 \\ x(1) &= 0 = c_3 + c_4 \\ c_1 t_1 + c_2 &= t_1^2 + 2 \\ c_3 t_1 + c_4 &= t_1^2 + 2 \end{aligned}\right\} \; c_1, c_2, c_3, c_4, t_1$$

t_1.

$$g + \frac{\partial g}{\partial \dot{x}}(\dot{\theta} - \dot{x})\Big|_{t_1^-} = g + \frac{\partial g}{\partial \dot{x}}(\dot{\theta} - \dot{x})\Big|_{t_1^+}$$

$\theta = t^2 + 2$

$\dot{\theta} = 2t$

$$\frac{2t_1 c_1 + 1}{(1+c_1^2)^{1/2}} = \frac{2t_1 c_3 + 1}{[1+c_3^2]^{1/2}} \longrightarrow t_1 = ?$$

3. $\dot{x} = u - u^2$

min time

$J = \int_0^{t_f} 1\, dt$ $\quad$ $H = g + \vec{\lambda}^T \vec{f} = 1 + \lambda(u - u^2)$

$\frac{\partial H}{\partial u} = 0 \Rightarrow \underbrace{\lambda}_{0\ 혹은}\underbrace{(1-2u)}_{0}$ $\quad u = \frac{1}{2}$ $\quad$ $\dot{\lambda} = -\frac{\partial H}{\partial x} \Rightarrow \lambda = \text{const}.$ $\quad \lambda \ne 0$

$H(t_f) = \frac{\partial H}{\partial t}\Big|_{t_f} = 0.$ $\quad$ $H = 0 \Rightarrow \lambda = -4$

반복하는 λ는 0일 수 없다.

$\dot{x} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ $\Rightarrow x = \frac{1}{4}t + c$ $\quad$ $x(0) = 1 \Rightarrow c = 1$

$x^* = \frac{1}{4}t + 1$

$x(t_f) = 0$ $\quad$ $0 = \frac{1}{4}t_f + 1$

$t_f = -4$

4. Ex. 5-2-3. Linear tracking.

$\int (x_1(t) - 1)^2$

$r(t) = 1$